

CSC420 Assignment 3

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Part I: Theoretical Problems

Question 1: RANSAC

1.1

Based on description, we have:

$$p = 99.5\% = 0.995$$

$$(1 - e) = 70\% = 0.7$$

$$s = 4$$

The number of iterations(samples) to fit the homography:

$$\begin{aligned} N &= \frac{\log(1 - p)}{\log(1 - (1 - e)^s)} \\ &= \frac{\log(1 - 0.995)}{\log(1 - 0.7^4)} \\ &= \frac{\log(0.005)}{\log(0.7599)} \\ &= \frac{-2.30102999566}{-0.11924355549} \\ &\approx 19.2969 \end{aligned}$$

Therefore, we need $\lceil N \rceil = \lceil 19.2969 \rceil = 20$ iterations to fit the homography.

1.2

Affine transformation requires 3 pairs of corresponding points, instead of 4 pairs. With a smaller s , the probability of having an all-inlier sample increases. Therefore N decreases, fewer RANSAC iterations are needed.

Question 2: Camera Models

2.1

Pixel coordinates of the line in the image:

$$\begin{aligned}\vec{p} &= \begin{bmatrix} wx \\ wy \\ w \end{bmatrix} = K \vec{P} \\ &= \begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} X_0 + td_x \\ Y_0 + td_y \\ Z_0 + td_z \end{bmatrix} \\ &= \begin{bmatrix} fX_0 + ftd_x + p_xZ_0 + p_xtd_z \\ fY_0 + ftd_y + p_yZ_0 + p_ytd_z \\ Z_0 + td_z \end{bmatrix}\end{aligned}$$

Vanishing coordinates of the line:

$$\begin{aligned}x &= \lim_{t \rightarrow \infty} \frac{fX_0 + ftd_x + p_xZ_0 + p_xtd_z}{Z_0 + td_z} = \frac{fd_x + p_xd_z}{d_z} = p_x + \frac{fd_x}{d_z} \\ y &= \lim_{t \rightarrow \infty} \frac{fY_0 + ftd_y + p_yZ_0 + p_ytd_z}{Z_0 + td_z} = \frac{fd_y + p_yd_z}{d_z} = p_y + \frac{fd_y}{d_z}\end{aligned}$$

We have the vanishing point for line L at:

$$(p_x + \frac{fd_x}{d_z}, p_y + \frac{fd_y}{d_z}).$$

2.2

Following hint, since the line lies on the plane, we have:

$$\vec{n} \cdot \vec{d} = n_x d_x + n_y d_y + n_z d_z = 0.$$

$$n_x \frac{d_x}{d_z} + n_y \frac{d_y}{d_z} + n_z = 0,$$

From Part 1, the vanishing point corresponding to a line with direction $\vec{d}$ is given by

$$(x, y) = \left(f \frac{d_x}{d_z} + p_x, f \frac{d_y}{d_z} + p_y \right).$$

we have:

$$\frac{d_x}{d_z} = \frac{x - p_x}{f}, \quad \frac{d_y}{d_z} = \frac{y - p_y}{f}.$$

Substituting these into the condition gives

$$n_x \frac{x - p_x}{f} + n_y \frac{y - p_y}{f} + n_z = 0.$$

Multiplying through by f results in:

$$n_x(x - p_x) + n_y(y - p_y) + n_z f = 0.$$

This equation represents a straight line in the image plane. Therefore, the vanishing points of all lines lying on the plane lie on this line horizon.

Question 3: Homogeneous Coordinates

3.1

Let the homogeneous coordinates of the two lines be:

$$l = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \quad l' = \begin{bmatrix} a' \\ b' \\ c' \end{bmatrix}.$$

point p via cross product:

$$p = l \times l' = \begin{vmatrix} i & j & k \\ a & b & c \\ a' & b' & c' \end{vmatrix} = \begin{bmatrix} bc' - cb' \\ ca' - ac' \\ ab' - ba' \end{bmatrix}.$$

To show that p is the intersection of l and l' , we need to verify that p lies on both lines:

$$\begin{aligned} l^T p &= a(bc' - cb') + b(ca' - ac') + c(ab' - ba') \\ &= abc' - acb' + bca' - bac' + cab' - cba' \\ &= 0 \end{aligned}$$

Similarly:

$$\begin{aligned} l'^T p &= a'(bc' - cb') + b'(ca' - ac') + c'(ab' - ba') \\ &= a'bc' - a'cb' + b'ca' - b'ac' + c'ab' - c'ba' \\ &= 0 \end{aligned}$$

Since p satisfies $l^T p = 0$ and $l'^T p = 0$, it lies on both lines l and l' . Therefore, the intersection of the two lines in homogeneous coordinates is given by: $p = l \times l'$.

3.2

Let the homogeneous coordinates of the two points be:

$$p = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, \quad p' = \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}.$$

line l via cross product:

$$l = p \times p' = \begin{vmatrix} i & j & k \\ x & y & 1 \\ x' & y' & 1 \end{vmatrix} = \begin{bmatrix} y - y' \\ x' - x \\ xy' - x'y \end{bmatrix}.$$

To show that l goes through p and p' :

$$\begin{aligned} l^T p &= (y - y')x + (x' - x)y + (xy' - x'y)1 \\ &= yx - y'x + x'y - xy + xy' - x'y \\ &= 0 \end{aligned}$$

Similarly:

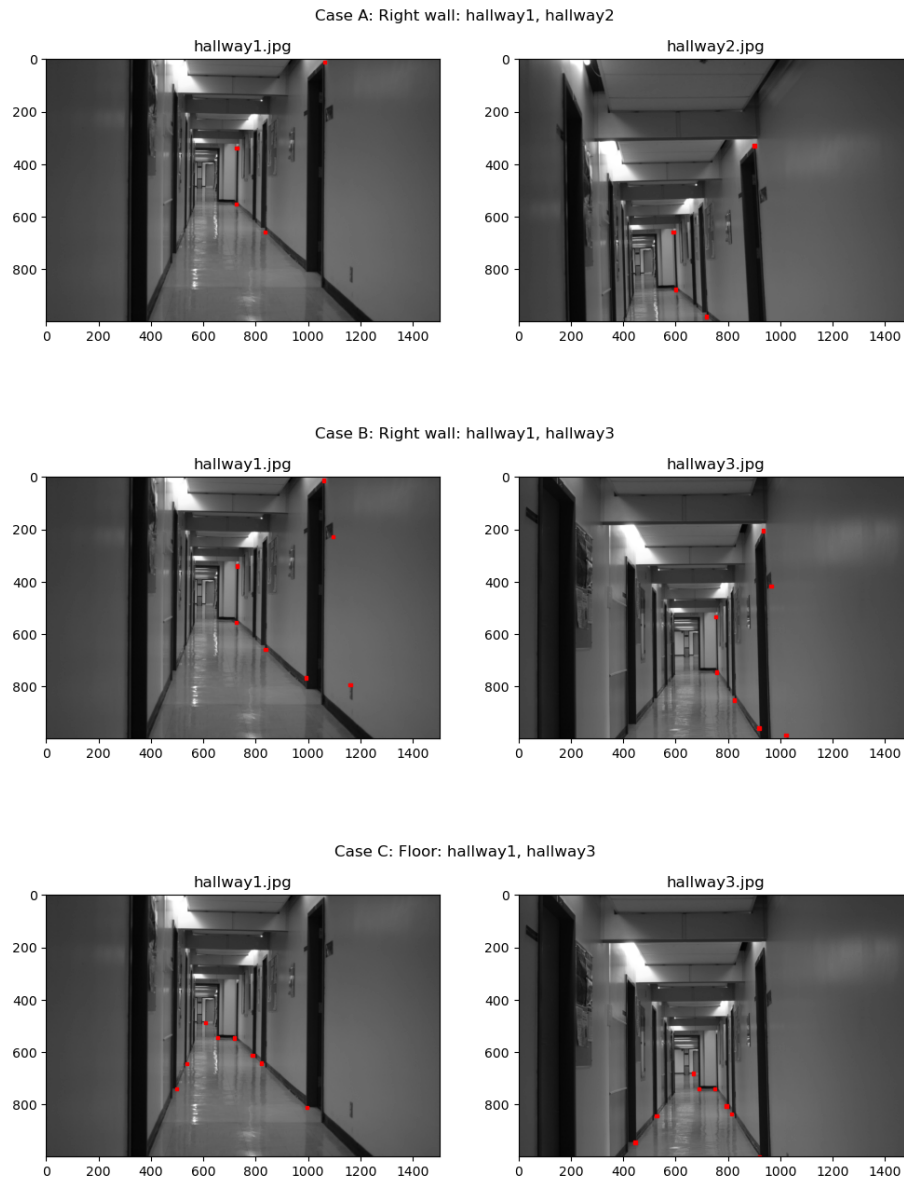
$$\begin{aligned} l'^T p' &= (y - y')x' + (x' - x)y' + (xy' - x'y)1 \\ &= yx' - y'x' + x'y' - xy' + xy' - x'y \\ &= 0 \end{aligned}$$

Since l satisfies $l^T p = 0$ and $l'^T p' = 0$, it goes through p and p' , and satisfies: $l = p \times p'$.

Part II: Implementation Tasks

Question 4: Homography

4.1 corresponding points

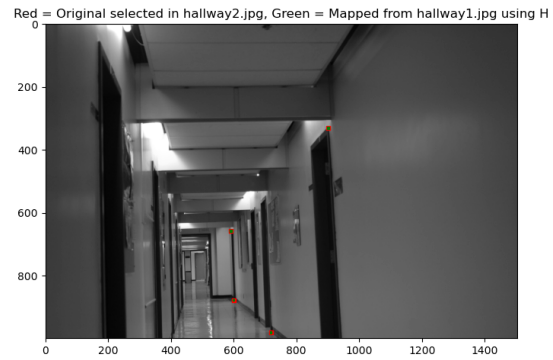


4.2 fitting homography

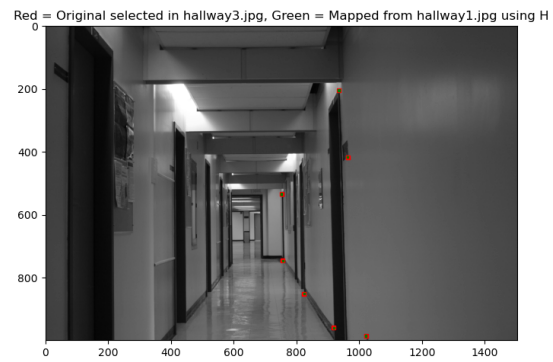
- Case A: Right wall: hallway1, hallway2
Estimated Homography (H) =
[[1.11418672e+00 6.45826835e-03 -1.95425538e+02]
[1.07535227e-02 1.00748626e+00 3.38631466e+02]
[9.21056763e-05 -6.81090744e-05 1.00000000e+00]]
Scaling: From entries (1,1) and (2,2), approximately 1.114 and 1.007, the image is slightly scaled up both horizontally and vertically.
Translation: From entries (1,3) and (2,3), the image is shifted $x = -195$ and $y = 339$ by large amounts.
Shear/Rotation: From entries (1,2) and (2,1), (0.00646, 0.01075), there is only a minimal shear (or rotation) effect.
- Case B: Right wall: hallway1, hallway3
Estimated Homography (H) =
[[5.34898952e-01 6.85763306e-03 3.35930226e+02]
[-3.33510849e-02 9.41727594e-01 2.20856776e+02]
[-3.25040521e-05 -2.51289448e-05 1.00000000e+00]]
Scaling: the image is scaled down horizontally by 0.535, and slightly down vertically by 0.942.
Translation: the image is shifted $x = 336$ and to $y = 221$.
Shear/Rotation: there is only a minimal shear or rotation effect: (0.00686, 0.03335).
- Case C: Floor: hallway1, hallway3
Estimated Homography (H) =
[[1.09817473e+00 -4.38647155e-01 2.32965909e+02]
[4.12180152e-02 1.02463157e+00 1.81945408e+02]
[7.87914388e-05 -2.84615181e-05 1.00000000e+00]]
Scaling: the image is scaled up horizontally by 1.098, and vertically by 1.025.
Translation: the image is shifted $x = 233$ and to $y = 182$.
Shear/Rotation: there is significant shear or rotation effect: (-0.438, 0.041), tilted counterclockwise.

4.3 mapping via homography

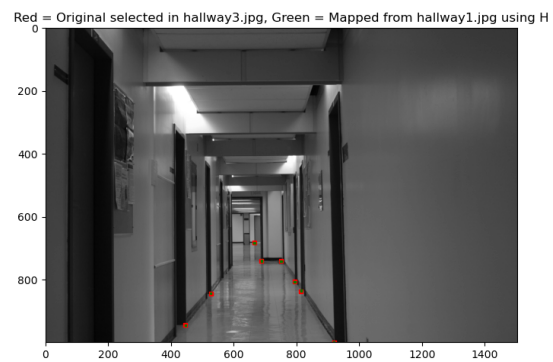
Case A: Right wall: hallway1, hallway2:



Case B: Right wall: hallway1, hallway3:

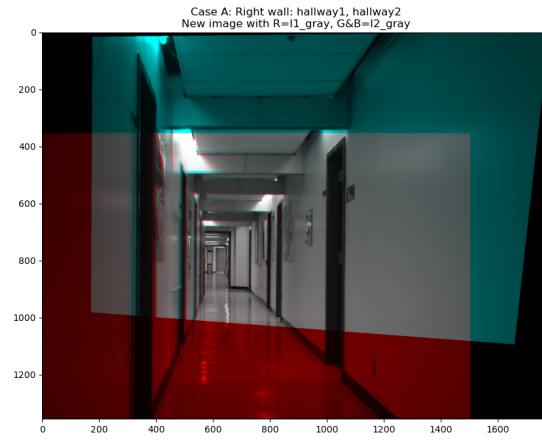


Case C: Floor: hallway1, hallway3:

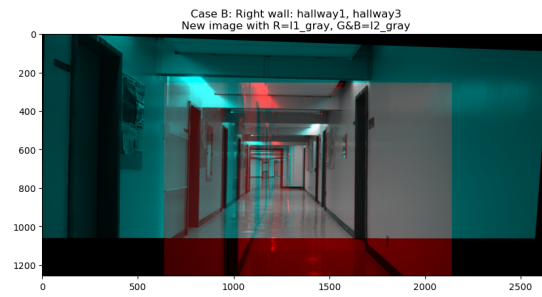


4.4 merged image

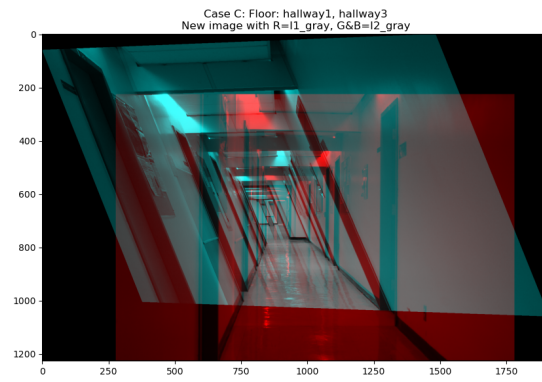
Case A: Right wall: hallway1, hallway2:



Case B: Right wall: hallway1, hallway3:



Case C: Floor: hallway1, hallway3:



Conclusion:

In Case A, hallway1 and hallway2 have similar 3D position but have different orientations, and the merged image show very little parallax within the shared region. However, in Cases B and C, hallway1 and hallway3 have very different 3D positions, and the merged image show very significant parallax. Therefore, parallax is dependent on the position of the camera instead of the viewing angle.

The right wall in Cases A and B shows a consistent gray in the shared regions, suggesting it is close to Lambertian. The floor in Case C, however, shows more variation, and therefore is less Lambertian.

4.5 Switch cases code

```
# Change to 'B' or 'C' as needed
# rerun entire Question 4 after change
case = 'A'

if case == 'A':
    case_description = "Right wall: hallway1, hallway2"
    img1_path = 'hallway1.jpg' # I1
    img2_path = 'hallway2.jpg' # I2
    pts1 = np.array([[727.17975096, 337.84330464],
                     [1062.7991877, 11.95196751],
                     [837.10727662, 659.84340192],
                     [726.206941, 552.83430614]], dtype=float)
    pts2 = np.array([[590.98635634, 657.89778199],
                     [901.31273408, 330.06082494],
                     [718.42446131, 980.87068924],
                     [599.74164599, 877.75283331]], dtype=float)
elif case == 'B':
    case_description = "Right wall: hallway1, hallway3"
    img1_path = 'hallway1.jpg'
    img2_path = 'hallway3.jpg'
    pts1 = np.array([[729.12537088, 340.76173452],
                     [1059.88075782, 12.92477747],
                     [1095.8747264, 228.88858894],
                     [1161.05299382, 793.11836665],
                     [992.75687047, 767.82530765],
                     [838.08008658, 658.87059195],
                     [726.206941, 554.77992607]], dtype=float)
    pts2 = np.array([[753.44561992, 534.35091687],
                     [934.38827278, 204.5683399],
                     [964.54538158, 417.61372148],
                     [1021.94116932, 985.73473904],
                     [918.82331339, 959.46887008],
                     [824.46074712, 852.45977431],
                     [756.36404981, 746.4234885]], dtype=float)
```

```

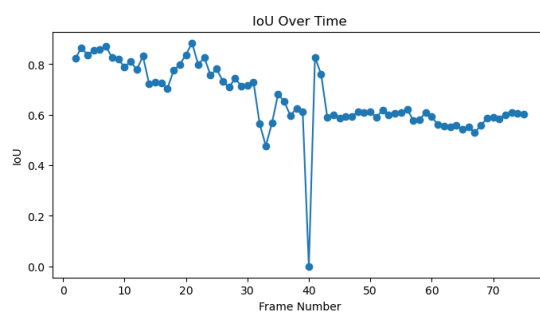
elif case == 'C':
    case_description = "Floor: hallway1, hallway3"
    img1_path = 'hallway1.jpg'
    img2_path = 'hallway3.jpg'
    pts1 = np.array([[609.46974561, 487.65603872],
                     [655.1918138, 545.05182645],
                     [718.42446131, 546.99744637],
                     [788.46677854, 613.14852376],
                     [821.54231723, 642.33282261],
                     [995.67530036, 812.57456588],
                     [497.59660003, 741.55943869],
                     [536.50899849, 644.27844253]], dtype=float)
    pts2 = np.array([[666.86553334, 682.21803103],
                     [688.2673525, 740.58662873],
                     [750.52719004, 741.55943869],
                     [794.30363831, 806.73770611],
                     [814.7326475, 836.89481492],
                     [919.79612335, 999.35407851],
                     [446.03767207, 944.87672066],
                     [526.78089888, 844.67729462]], dtype=float)

```

Question 5: Mean Shift Tracking

5.1 Performance Evaluation

IoU over time:



Sample frames:

High IoU: 88.45% > 50%

Low IoU: 0% < 10%, where Viola-Jones failed to detect the face.

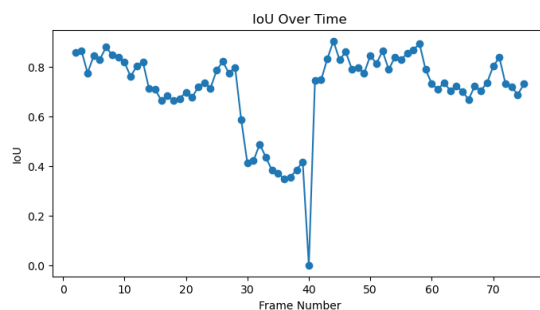


The percentage of frames in which the IoU is larger than 50%: 97.30%.

In the frame where the IoU is below 10%, the Mean-Shift tracked bounding box appears to be correct more often than the Viola-Jones detection. This is likely because Viola-Jones fails when the face is partially blocked or changes orientation, while the Mean-Shift tracker remains locked onto the color distribution it learned from the initial frame.

5.2 Implement a Simple Variation

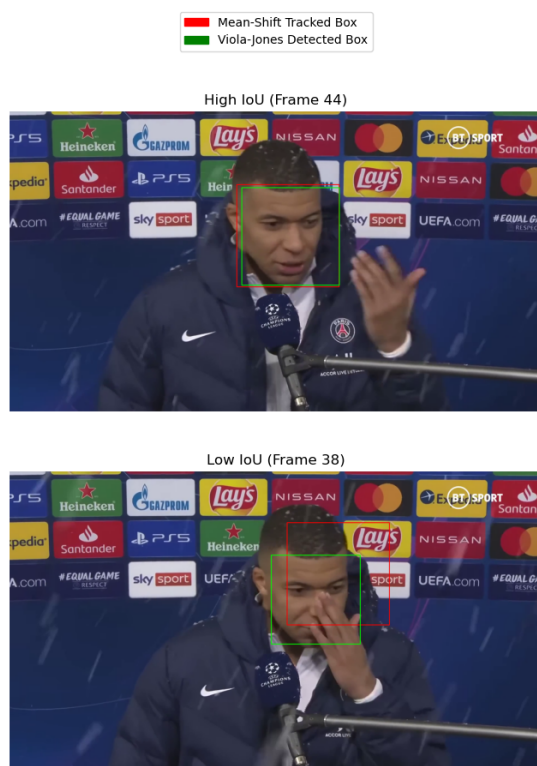
IoU over time:



Sample frames:

High IoU: 90.48% > 80%

Low IoU: 38.42% < 50%. It's worth noting that the lowest IoU is still 0%, where Viola-Jones failed to detect the face. Here, I picked another non-zero low IoU.



The percentage of frames in which the IoU is larger than 50%: 85.14%.